Machine Learning in Quantitative Finance — Project Guide

Introduction

Machine Learning is transforming quantitative finance by introducing adaptive, data-driven approaches to trading, risk management, and portfolio construction.
This document provides **three advanced, research-oriented projects** that combine machine learning techniques with established financial concepts.
Each project includes background, implementation goals, evaluation methods, and optional extensions for deeper exploration. Also there's a gift for everyone who makes it through this document at the end :)

Topic 1: Improving Basket Trading Using Bayesian Optimization

Overview

Basket trading strategies rely on identifying cointegrated relationships among assets to exploit temporary price deviations.
Traditional approaches, such as the **Johansen test**, generate in-sample cointegrating weights, but these often fail to generalize out-of-sample.
This project explores the use of **Bayesian Optimization (BO)** — a global optimization technique — to improve cointegration-based basket trading performance.

Core Idea

Bayesian Optimization efficiently searches for optimal cointegrating weights by modeling the trading performance as a black-box function.
The goal is to find parameter configurations that maximize out-of-sample profitability or stability rather than relying solely on statistical p-values.

Implementation Steps

1. Identify a set of cointegrated assets (e.g., equity pairs, sector ETFs, currency baskets).
2. Apply the Johansen test to obtain baseline cointegrating weights.
3. Implement Bayesian Optimization to search for weights that maximize a custom objective (e.g., Sharpe ratio, mean return / volatility ratio).

4. Evaluate using a **rolling window approach** to assess out-of-sample performance.

Evaluation

- Rolling window performance across multiple time periods.
- Key metrics: **Sharpe ratio**, **drawdown**, **profit factor**, **mean reversion half-life**.
- Compare BO-optimized strategy against Johansen baseline.

Optional Extensions

- Tune trading thresholds (entry/exit bands) via BO as well.
- Implement **multi-objective BO** to optimize for multiple criteria (e.g., return vs. volatility).
- Incorporate realistic **transaction costs** and **slippage** models.

Topic 2: Robust Portfolio Optimization via Smart Predict-then-Optimize (SPO)

Overview

In traditional portfolio optimization, we first predict asset returns and covariance matrices, then feed these into an optimization model — a **two-stage process**.
However, prediction accuracy doesn't always translate into better portfolio decisions.
The **Smart Predict-then-Optimize (SPO)** framework integrates prediction and optimization into a **single end-to-end learning model** that minimizes actual decision loss.

Core Idea

Instead of minimizing prediction error, SPO directly trains the predictive model (e.g., neural network, regression, tree model) to minimize portfolio loss — aligning learning objectives with downstream optimization outcomes.

Implementation Steps

1. Choose a portfolio optimization model (e.g., **mean-variance** or **mean-CVaR**).
2. Build a predictive model for returns and covariances.

3. Embed the optimization layer within the learning framework (SPO).

4. Train end-to-end to minimize the **decision loss** rather than pure prediction loss.

Evaluation

- Compare **SPO** vs. **two-stage (predict → optimize)** approaches.
- Use standard metrics: **portfolio return**, **volatility**, **Sharpe ratio**, **CVaR**.
- Test under different market regimes (bull, bear, sideways).

Optional Extensions

- Incorporate **distributionally robust optimization (DRO)** in the portfolio layer.
- Experiment with **different loss weightings** between prediction and decision error.
- Add **transaction cost** and **turnover penalties** for realistic portfolios.

Topic 3: Derivative Hedging Using Reinforcement Learning

Overview

Traditional hedging approaches like **delta** or **gamma hedging** assume simplified market dynamics and static parameters.
In practice, market conditions evolve dynamically.
This project applies **Reinforcement Learning (RL)** to develop **adaptive hedging strategies** that learn to minimize portfolio risk in real-time.

Core Idea

Use RL agents to dynamically adjust hedge ratios (e.g., delta, gamma, vega) in response to changing market states.
The model continuously updates decisions based on simulated or historical price data, volatility shifts, and transaction costs.

Implementation Steps

1. Define **state space** (e.g., current option Greeks, underlying price, time to maturity).

2. Define **action space** (e.g., buy/sell hedge units).

3. Design a **reward function** that penalizes large P&L drawdowns and transaction costs.

4. Train the agent (e.g., DQN, PPO, or SAC) on rolling historical windows.

5. Compare results to traditional **delta** and **gamma hedging** benchmarks.

Evaluation

- Metrics: **Hedging error**, **portfolio variance**, **transaction cost efficiency**, **drawdown**.

- Compare RL-based strategies with static or rule-based baselines.

Optional Extensions

- Implement **Distributionally Robust RL (DRRL)** to hedge against model uncertainty.

- Test **alternative reward structures** (e.g., asymmetric penalties for losses).

- Explore **cost-aware** or **risk-sensitive** RL variants.

Suggested Tools & Libraries

- **Python Libraries:** NumPy, Pandas, scikit-learn, PyTorch / TensorFlow, Optuna, Ax-Platform (for BO).

- **Finance Libraries:** yfinance, QuantStats, PyPortfolioOpt, FinRL.

- **Datasets:**

    - Yahoo Finance (equities, ETFs, FX)

    - Quandl (macro & derivatives data)

    - Kaggle Quant datasets (e.g., JPX, G-Research)

    - OptionMetrics or synthetic simulation (for hedging tasks)

Feeling Overwhelmed? Start With the Fundamentals

If these projects feel a bit advanced or you're still new to programming then don't worry — everyone starts somewhere.
To help you build all the skills you need for ML and quantitative finance from the ground up, I created a complete **Data Science Roadmap** that walks you through Python, SQL, Math/Stats, Machine Learning and more, with resources and beginner-friendly projects for every chapter.

Because you made it to the end of this document you can get an **additional 25% off** the roadmap using the code **QUANT**. Join 20+ students who are already inside the Roadmap today and get started with your coding journey.